27 An experiment is set up to demonstrate the diffraction of water waves in a ripple tank.

The waves pass through a gap of width *w* and some diffraction of the waves is observed.

The wavelength of the waves is now doubled.

What is the new gap width needed to cause the same amount of diffraction as before?

A 0.5*w* **B** *w* **C** 2*w* **D** 4*w*